



**NATIONAL COMPETENCY STANDARDS &
COMPETENCY BASED TRAINING CURRICULUM
FOR
INFORMATION AND COMMUNICATION TECHNOLOGY
TECHNICIAN**

(Competency Standard Code: K72S004)



**Developed and Validated by;
National Apprentice & Industrial Training Authority
971, Sri Jayawardenepura Mawatha,
Welikada,
Rajagiriya.**



**Endorsed by;
Tertiary & Vocational Education Commission
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**Ministry of Education
Skill Development and Vocational Education Division**

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PREFACE

National Competency Standards (NCS) and curricula are essential elements for the implementation of a unified Technical and Vocational Education & Training System for Sri Lanka and forms the basis for the National Vocational Qualifications Framework of Sri Lanka, which provides for the award of nationally and internationally recognized qualifications. The National Competency Standards are developed in consultation with the industry and are designed using a nationally agreed specific format to maintain uniformity and consistency of standards amongst occupations.

The NCSs and curricula specify the standards of performance of a competent worker and the various contexts in which work may take place. NCS describes the knowledge, skills and attitudes required in a particular occupation and signals what students must be able to do in employment and can ideally serve as a link between education and training and the needs of the labour market. NCS and curricula together provide explicit advice to assessors and employers regarding the knowledge, skills and attitudes to be demonstrated by the candidates seeking formal recognition for the competencies acquired either following training or through work experience.

National Competency Standards focus on that is expected of a worker in the workplace rather than on the learning process.

- NCSs stress upon the ability to transfer and apply knowledge and skills to new situations and environments.
- In NCS and curricula the emphases are upon outcomes and upon the application of knowledge and skills, not just the specification of knowledge and skills.
- NCS and curricula are concerned with what people are able to do and the ability to do a task in a range of contexts.
- NCS and curricula include all aspects of workplace performance and not only narrow task skills.

NCS especially can serve a number of purposes including;

- Providing advice to curriculum developers about the knowledge, skills and attitude to be included in the curriculum.
- Providing specifications to Competency Based Assessors about the knowledge, skills and attitudes to be demonstrated by candidates.
- Providing advice to industry about job functions, which in turn can be used for the development of job descriptions, performance appraisal systems and work flow analysis.

The lead organization for the development of NCS and curricula is the National Apprentice & Industrial Training Authority. The standards so developed are endorsed by the Tertiary & Vocational Education Commission as National Document.

There is a requirement to review the standards and curricula within the prescribed period as appropriate as and when required, with the assistance of relevant industry groups and incorporate the changes in the National Competency Standards and Curricula.

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**NATIONAL COMPETENCY STANDARD & COMPETENCY BASED TRAINING
CURRICULUM
FOR
INFORMATION AND COMMUNICATION TECHNOLOGY
TECHNICIAN**

(Competency Standards Code: K72S004)

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**NATIONAL COMPETENCY STANDARDS
FOR THE OCCUPATION OF
INFORMATION AND COMMUNICATION TECHNOLOGY TECHNICIAN
NVQ Level 04**

	List of Competency Units	Unit Code No.	Level	Page No
1.	Use the computer and manage files within standard operating systems	K72S004U01	2	01-03
2.	Perform word processing	K72S004U02	2	04-07
3.	Prepare spreadsheets	K72S004U03	3	08-11
4.	Prepare presentation resources	K72S004U04	3	12-14
5.	Perform basic operations of cloud computing including internet and email services	K72S004U05	2	15-17
6.	Handle relational database	K72S004U06	3	18-20
7.	Conduct routine maintenance services of computer system and peripherals	K72S004U07	3	21-22
8.	Produce print page layout and multimedia objects	K72S004U08	4	23-24
9.	Analyze, design and develop information system	K72S004U09	4	25-26
10.	Handle relational databases using standard SQL	K72S004U10	4	27-28
11.	Design static website	K72S004U11	4	29-30
12.	Practice workplace communication and interpersonal relations	K72S004BU01	Basic	31-32
13.	Apply occupational literacy and numeracy	K72S004BU02	Basic	33-34
14.	Work in a team	K72S004BU03	Basic	35-36
15.	Practice occupational health and safety measures	K72S004BU04	Basic	37-40

Packaging for National Vocational Qualifications (NVQ)

- National Certificate at **NVQ Level 2** in the occupation of **Data Entry Operator** will be awarded to those who are competent in U01 + U02 + U05 + BU01 + BU02 + BU03 + BU04 – **K72S004Q1L2**
- National Certificate at **NVQ Level 3** in the occupation of **Computer Application Assistant** will be awarded to those who are competent in U01 + U02 + U03 + U04 + U05 + U06 + U07 + BU01 + BU02 + BU03 + BU04 – **K72S004Q2L3**
- National Certificate at **NVQ Level 4** in the occupation of **Information and Communication Technology Technician** will be awarded to those who are competent in U01 + U02 + U03 + U04 + U05 + U06 + U07 + U08 + U09 + U10 + U11 + BU01 + BU02 + BU03 + BU04 – **K72S004Q3L4**

Occupational Outlook

Information and Communication Technology Technician

Job Description

Information and Communication Technology Technician is a person who is competent in preparing well formatted documents which include analyzed and summarized data in graphical format. He/she should be able to provide technical support for both hardware and software for the smooth operation of the organization. When required designing web sites and desktop applications with database connectivity is also a responsibility of the ICT Technician.

Responsibilities

ICT Technician is responsible to carry out his/her duties which fulfill the customer requirement while adhering to quality and security standards enforced by the relevant organizations.

- prepare well formatted reports and related documents
- prepare spreadsheets and charts which can be used for data analysis
- build databases to store and manage data in the organization as and when required
- handle emails and cloud services
- handle the operations of computer systems and related devices
- troubleshoot computer systems, network and related devices
- perform routine maintenance of computer systems, network and related devices
- install new software as and when required
- update the existing software as and when required (specially operating system and antivirus software)
- design/prepare simple webpages/websites
- develop and deploy desktop application with database connectivity

Career and Employment Opportunities

Those who have completed this competency level in the occupation of ICT Technician have many employment opportunities such as, Computer Operator, Data Entry Operator, Computer Technician, Network Assistant, Computer Programmer, Web Designer, and Database Assistant in national and international organizations.

They can be entrepreneurs and self-employees depending on their wish and dedication.

They have opportunities to continue their higher education and obtain higher vocational qualification in the field of Computer Hardware, Computer Networking, Software Development, Web Design & Development, Software Quality Assurance and Computer Security.

Career paths include, but are not limited to:

- | | |
|--------------------------|-----------------------------|
| • Computer Programmer | • Quality Assurance Officer |
| • Software Developer | • System Designer |
| • Web Designer | • System Analyst |
| • Web Developer | • Software Engineer |
| • Database Administrator | • Hardware Engineer |
| • Hardware Technician | • Network Engineer |
| • Network Administrator | • Software Architect |
| • Software Tester | • Project Manager |

Unit Title:	Handle computer and files within standard operating systems
Unit Descriptor:	This unit covers the competencies required to perform basic configuration of the computer, all file / folder functions using the operating system, maintain system using system tools while adhering to safe operations of device and data.
Unit Code:	K72S004U01
Elements of Competence	Performance Criteria
1. Start up and shutdown computer	1.1 Computer started up in the following standard startup procedure, starting by switching on with the protected power supply 1.2 Computer shut down in the following standard shutdown procedure after terminating the applications in the correct order 1.3 Power option properties changed as per the standard procedure 1.4 Computer restarted using a standard routine if required. 1.5 A non-responding application terminated as specified.
2. Perform basic configuration	2.1. System information, operating system and version number identified. 2.2. Desktop configuration of the computer, date and time display system, color settings etc. changed according to given instructions 2.3. Keyboard language set according to given instructions. 2.4. Display settings changed as per the standard procedure 2.5. Application software installed and uninstalled as per the standard procedure. 2.6. Fonts installed according to given instructions. 2.7. Regional setting changed according to given instructions 2.8. Help functions used as appropriate
3. Maintain system using system tools.	3.1. Disk errors checked when required 3.2. Backups taken as per the standard procedure 3.3. Computer local storage managed 3.4. Recycle bin storage managed
4. Perform text editing system	4.1. A text editing application launched, a file opened, a new file created, named and placed as required. 4.2. File saved to a given location on a given drive and named as required and text editing application closed.
5. Work with Icons on desktop	5.1. Desktop icons identified selected and moved as required. 5.2. File, directory / folder and applications opened from the desktop. 5.3. Desktop icon/s and menu shortcut created as required.

6. Work with windows	6.1 Window minimized, maximized, resized, moved and closed by clicking on the appropriate symbols available on the tool bars. 6.2 Tiling, cascading and splitting windows, and switching between different open windows, performed as required.
7. Manage files and a folder	7.1 Usage of hierarchical file system containing drives, folders, sub-folders and files demonstrated. 7.2 A directory / folder and sub-folder, created in storage devices as required. 7.3 Common file types applied as per the standard procedure. 7.4 File and folder attributes such as archive, read-only, etc. changed as required. 7.5 Selection of files / folders either individually or as adjacent or non-adjacent groups, renaming, coping and moving of files / folders performed using the required options. 7.6 Deletion and recovery of files / folders to and from recycle bin / waste basket performed using the required options. 7.7 Files / folders searched using find / search tool, by content, date modified, wildcards as required. 7.8 Files compressed and extracted as required and procedures.
8. Manage antivirus software.	8.1 Antivirus software settings applied as per the standard procedure. 8.2 Specified drives, folders and files scanned using current version of a virus scanning applications. 8.3 Scanning software / virus definition updated regularly.

Range Statement:

Work connected to this unit may take place in an office environment or in a computer laboratory using standard operating system. work related to this unit may be performed individually or as a member of team.

Elaboration of Terms in Performance Criteria

- **Storage devices** in PC 7.2 may include: Pen drives, external hard disc, Data tours, Memory chips
- **Common file types** in PC 7.3 may include such as word processing files, spreadsheet files, data base files, presentation files, image files, audio, video and PDF files

Critical aspects

The assessment must confirm that the candidate is able to;

- Operate a computer with standard operating system accurately.
- Perform all file / folder functions using the operating system accurately
- Ensure safe operations of device and data

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Personal computer	• Standard operation system • Software applications. • External storage media • Anti-Virus software on CD

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Intellectual Property Act No. 36 of 2003
- Windows 11 All-in-One for Dummies, Ciprian Adrian Rusen, Publisher: Wiley, 2022 ISBN :1119858712, 9781119858713

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• English literacy. • Basic knowledge of computer configurations and operations, main components of computers, their functions and mode of operations. • Standard operating systems. • Basics of computer viruses and spyware other harmful application (malware) and common methods of protection.	• Connect relevant external parts / peripherals of computers. • Use mouse and keyboard

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Perform word processing
Unit Descriptor:	This unit covers the competencies required to use a word processing application in a computer, create, format and print documents containing text, tables, images, and use special features such as “mail merge”
Unit Code:	K72S004U02
Element	Performance Criteria
1. Use a word processing application	1.1 New documents created, based on the default template and appropriate template. 1.2 Document saved to a designated location under its existing name or a new name with or without password as required. 1.3 Document saved in an appropriate file format. 1.4 Saved documents opened and closed as appropriate. 1.5 HELP functions used as appropriate.
2. Perform adjustment of word processor settings	2.1. Display /hiding of toolbars ruler and non-printing characters demonstrated correctly. 2.2. Basic options / preferences in the application such as file security user information, default file location modified as required. 2.3. Auto correction options setup as required. 2.4. Auto save function is enabled/disabled as required 2.5. Language preference setup as required
3. Perform text processing operations	3.1 Text typed, special characters, Auto text entries and symbols inserted following accepted principles in text processing as required. 3.2 Paragraph marks and line break marks inserted and removed as required. 3.3 A single character, word, line, sentence, paragraph or entire body of text selected as required. 3.4 Content edited as required by inserting new characters/words into existing text, replacing existing text 3.5 Text copied/moved as required within a document or between documents and unnecessary text deleted. 3.6 Words and / or phrases replaced as required, using the find and replace commands.
4. Format text and paragraph	4.1. Text appearance changed as required by changing character attributes 4.2. Character spacing adjusted as required. 4.3. Text alignment, first line indentation line spacing within paragraphs and spacing above and below paragraphs changed as required. 4.4. Tab indents set and removed. 4.5. Styles defined, applied and formatting transferred from one piece of text to another 4.6. Different styles of bullets and numbers applied or removed from single and multiple level lists as required. 4.7. Borders and shading added to a paragraph

5. Format a document	5.1. Different number of columns applied to the document as required. 5.2. Document orientation, paper size, and margins changed as required 5.3. Page break, section break inserted and deleted as required. 5.4. Header & footer functions and text added & removed as required. 5.5. Automatic page numbering applied to a document in the required format. 5.6. Page borders inserted and removed as appropriate.
6. Create and format tables in a document	6.1 Table created, data inserted and edited as required. 6.2 Rows, columns, cells or the entire table selected, inserted or deleted, column width and row height modified as required. 6.3 Cells merged and split as required. 6.4 Cell border width, style, color modified, shading added and text direction changed as required. 6.5 Table properties applied as required. 6.6 Data sorted and formula used for calculations as required.
7. Insert objects and link in to a document	7.1 Picture / image and chart inserted or linked into a document using Object Linking and Embedding (OLE) tools, selected, and duplicated within the document as required. 7.2 Picture, an image and a chart moved within a document and into another document as required. 7.3 A picture, an image and a chart resized and deleted as required. 7.4 Auto shape, word art, text boxes applied, and multiple objects managed (group, order, align etc.) as required. 7.5 Text wrapping applied as required.
8. Carryout mail merging of a document	8.1 Mailing list or other data file prepared for mail merge. 8.2 Main document prepared for mail merge by inserting data fields as required. 8.3 Data source merged with a main document such as a letter so that multiple copies are obtained correctly.
09.Review document	9.1 Track changes applied to the document 9.2 Comments inserted as required 9.3 Markup accepted/rejected as required
10.Prepare final document	10.1 Document proofed by checking layout, presentation, spelling etc. so that a correct final document is obtained. 10.2 Spelling and grammatical errors corrected using spell-check function. 10.3 Commonly used words added to a custom dictionary to facilitate spell checking on future occasions 10.4 Table of content (TOC) and Table of Figures generated as required 10.5 Document previewed using the preview function in order to identify formatting errors etc.

11. Save and print	11.1 Document saved under different <i>file format/designated location</i> as required 11.2 Suitable print options such as entire document, a selected section, specified pages, multiple copies, collate pages etc. selected as required. 11.3 Document printed from an appropriate printer using the required options.
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Range Statement:

The work outlined in this unit shall be demonstrated individually in an office environment or in a computer laboratory, and shall include all those competencies required to word process a document and use the functions available in standard word processing software.

Elaboration of Terms in Performance Criteria

- ***designated location*** in PC 1.2 & 11.1 may include file saving locations defined by the organization
- ***File format*** in PC 1.3 & 11.1: may include rich text format, PDF, HTML etc.
- ***accepted principles*** in PC 3.1 may include avoiding unnecessary text operations such as multiple space between words, multiple paragraph brake, multiple tab insert, etc.
- ***character attributes*** in PC 4.1 may include font types and sizes, and font style, underlined, subscript, superscript, case change, text effects, text color etc.

Critical aspects

The assessment must confirm that the candidate is able to;

- Setup the page setting as required
- Perform text processing and paragraph & document formatting
- Perform save and print management functions

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Computer with standard operating system and word process application • Printer	• Paper / print media • Printing cartridge for printer

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Intellectual Property Act No. 36 of 2003
- Shelly Cashman Series Microsoft Office 365 & Office 2019 Introductory (MindTap Course List) 1st Edition, by Sandra Cable (Author), Steven M. Freund (Author), Ellen Monk (Author), Susan L. Sebok (Author), Joy L. Starks (Author), ISBN: 0357026438, 978-0357026434

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• English literacy. • computer literacy • Health and safety procedures	• Ability to setup and use a computer and perform file management functions • Ability to type in multiple languages

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Prepare Spreadsheets
Unit Descriptor:	This unit covers the competencies required to work with standard spreadsheet programs, import and export data, set up simple formulae to perform calculations, create charts, simple data base management and obtain printouts of spreadsheets.
Unit Code:	K72S004U03
Element	Performance Criteria
1. Perform basic spreadsheet operations	1.1 A new spreadsheet created using the default template. 1.2 A spreadsheet saved to a designated location on a drive. 1.3 Switching between worksheets, renaming worksheets, copy and moving worksheets performed as required 1.4 New worksheet inserted renamed moved, copied and deleted as required. 1.5 Multiple work sheet selected as required. 1.6 Multiple worksheets' view arranged as required 1.7 Worksheet protected as required. 1.8 Help functions used as appropriate.
2. Adjust settings	2.1. Magnification /zoom tools used as required. 2.2. Built in toolbars displayed/hidden as required. 2.3. Row and / or column titles frozen / unfrozen as required. 2.4. Row and / or column hide / unhide as required 2.5. Basic options / preferences such as username, default directory / folder modified as required Cell size adjusted as required 2.6. Alignments set as required 2.7. Add/ remove styles (conditional formatting etc.) as required
3. Select, insert and delete cells, rows and column	3.1 A single cell, a range of adjacent cells, a range of non-adjacent cells, entire sheet selected as required. 3.2 A single row, a range of adjacent rows, a range of non-adjacent rows selected as required. 3.3 A single column, a range of adjacent columns, range of non-adjacent columns selected as required. 3.4 Rows and columns inserted in worksheet at the required place. 3.5 Rows and columns deleted without affecting the rest of the worksheet.

4. Format cells	4.1. Cells formatted to display number to a specific number currency, data, text and percentages 4.2. Cell content properties, font color, font sizes, font styles, background formatted as required, 4.3. Formatting copied from a cell or cell range to another cell or cell range as required. 4.4. Text wrapping applied; multiple lines inserted to cell contents as required. 4.5. Contents aligned orientation adjusted as required. 4.6. Border effects added to a cell or cell range as required.
5. Insert data / tables to spreadsheet	5.1. Data entered to spreadsheet, 5.2. Tables/ pivot tables inserted to spreadsheet 5.3. Cell contents edited as required 5.4. Data imported from other file formats to spreadsheet
6. Carry out Copying moving and deleting cell content/cell range	6.1 The contents of a cell, a cell range within a worksheet, copied/moved between worksheets and other open workbook. 6.2 Transpose and other pasting options applied while copying the data as required. 6.3 Data copied and incremented using the auto fill and copy handle tools. 6.4 Cell contents deleted without affecting other cells.
7. Search and replace cell content	7.1 Specific content in a worksheet located using the search command. 7.2 Specific content in a worksheet replaced using the replace command.
8. Sort and filtering data	8.1 A cell range sorted according to criterion in ascending / descending numerical and alphabetical order. 8.2 Data filtered as required 8.3 Conditional formatting used as required
9. Use arithmetic formulae and functions.	9.1 Relative, mixed, absolute, cell referencing in formulae carried out as required. 9.2 Formulae generated using cell references and arithmetic operators avoiding error values associated with formulae. 9.3 Formulae generated using sum, average, minimum, maximum and count functions. 9.4 Formulae generated using the logical function IF, yielding two specific values. 9.5 Define name applied for a cell

10. Create charts / graphs	10.1 Different types of charts/graphs such as column, bar, line, pie, created from spreadsheet data as required. 10.2 Titles / labels added / removed to /from a chart / graph as required. 10.3 The background color and colors of chart/graph changed as required. 10.4 The chart / graph type changed as required. 10.5 Charts / graphs copied / moved / re sized and deleted within a worksheet, between worksheets and between open spreadsheets as required.
11. Setup worksheet	11.1 Worksheet margins, orientation, paper size, page fit, changed as required. 11.2 Page breaks applied as necessary. 11.3 Header & footer properties changed as required 11.4 Print order adjusted as required.
12. Prepare and print spreadsheet	12.1 Spreadsheet calculation and text checked before printing to detect errors. 12.2 A worksheet previewed to check formatting etc. before numbering. 12.3 Display of grid lines, display of row and column headings etc. turned off prior to printing as required. 12.4 Automatic title row printing on every page applied as required. 12.5 A cell ranges from a worksheet or entire worksheet, the entire spreadsheet, or a selected chart printed as required.
13. Save and Export data	13.1 Spreadsheet saved under different <i>file format/designated location</i> as required 13.2 Data exported to other formats as required

Range Statement:

The work outlined in this unit shall be demonstrated individually in an office environment or in a computer laboratory and shall include all those competencies required to originate and edit spreadsheets, enter data and process data using the functions available in standard spreadsheet applications.

Elaboration of Terms in Performance Criteria

- ***designated location*** in PC 1.2 & 13.1 may include file saving locations defined by the organization
- ***File format*** in PC 1.3 & 13.1: may include rich text format, PDF, HTML etc.

Critical aspects

The assessment must confirm that the candidate is able to;

- Insert data in appropriate format and cell formatting
- Perform calculations using formulae & functions accurately.

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Personal computer with standard operating system and relevant applications software. • Printer	• Paper • Printing cartridge for printer

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Payment Devices Frauds Act No. 30 of 2006.
- Intellectual Property Act No. 36 of 2003
- Shelly Cashman Series Microsoft Office 365 & Office 2019 Introductory (MindTap Course List) 1st Edition, by Sandra Cable (Author), Steven M. Freund (Author), Ellen Monk (Author), Susan L. Sebok (Author), Joy L. Starks (Author), ISBN: 0357026438, 978-0357026434

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• English literacy. • Basic knowledge of computer configuration and operations, main components of computers, their functions and mode of operations. • Standard operating systems. • Arithmetic and logical operations • Health and safety procedures	• Ability to connect relevant external parts and peripherals of a computer. • Ability to power on and boot up a computer

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Prepare presentation resources
Unit Descriptor:	This unit covers the competencies required to create, format and modify presentations using various features available in the presentation software adhering to health and safety procedures.
Unit Code:	K72S004U04
Element	Performance Criteria
1. Perform basic operations with presentations software	1.1 Presentation application opened and closed properly. 1.2 Several presentations opened simultaneously. 1.3 New presentation created using the default template. 1.4 Presentation saved to a specified location on a drive. 1.5 Presentation saved under another name/location. 1.6 File exported as required
2. Adjust settings	2.1 Presentation setup according to the output device 2.2 Magnification / zoom tools used as required. 2.3 Ribbons/Toolbars displayed / hidden as required. 2.4 Basic options / preferences in the application such as user name, default directory / folder changed as required.
3. Change views	3.1. Different presentation view modes selected to suit the presentation. 3.2. Presentation view modes changed from one to another as required.
4. Add/Delete/Move slides	4.1. New slides with a specific layout added as required. 4.2. Changing from one slide layout to another carried out without affecting the contents. 4.3. Unwanted slides deleted /hidden 4.4. The slides rearranged by moving as required
5. Use design templates	5.1. Design template selected appropriately 5.2. Design templates modified as required.
6. Use slide masters	6.1 Slide masters selected appropriately 6.2 Slide master setup as required to accommodate slide numbers; update date/non update date etc.

7. Format text input	7.1 Text added into a presentation in standard / outline view as required. 7.2 Slide contents and notes page contents edited by inserting new characters /words as required. 7.3 Text appearance changed as required by changing character attributes 7.4 Decorative text applied as required. 7.5 Paragraph appearance changed as required by changing paragraph attributes as required. 7.6 Line spacing before and after bulleted or numbered points adjusted as required. 7.7 The style of bulleted or numbered lists changed as required. 7.8 Notes for the presenter added to slide as required.
8. Insert multimedia objects	8.1 An image inserted into a slide at the required place. 8.2 Images resized and edited as required. 8.3 External objects inserted as required 8.4 Chart inserted/modified as required
9. Copy, move and delete text and images	9.1 Text images copied/ moved within a presentation and between presentations as required. 9.2 Text images, resized and deleted as required
10. Insert/Modified smart art	10.1 Smart art inserted as required 10.2 Smart art formatted/modified as required
11. Insert/modify drawing objects into slides	11.1 Different types of drawing objects inserted into a slide as required. 11.2 Inserted drawing object formatted 11.3 Insert drawing object modified
12. Insert links	12.1 Hyperlink inserted (URL or book mark) as required 12.2 Action buttons inserted as required.
13. Apply transition effects, and animation	13.1 Custom animation effects , timing applied to slide objects as required 13.2 Transition effects between slides and advanced timing changed as required.
14. Prepare for presentation and/or printing	14.1 Presentation time recorded as required. 14.2 Presentation recorded as required. 14.3 Custom slide show created as required 14.4 A slide show presented starting from the beginning or a from a required slide 14.5 Presentation exported as required format

Range Statement:

The work outlined in this unit shall be demonstrated individually in an office environment or in a computer laboratory and shall include all those competencies required to prepare, edit, format and conduct presentations using a standard presentation software.

Elaboration of Terms in Performance Criteria

- **output device** in PC 2.1; may include on screen HD TV, smart board, printer
- **character attributes** in PC 7.3: may include font types and sizes, and font style, underlined, superscript, subscript, case change, text effects, text color etc.
- **Paragraph attributes** in PC 7.5: may include text alignment, Paragraph spacing and line spacing
- **External objects** in PC 8.3; may include file created with word document, Excel worksheet, paint brush picture, etc.
- **Smart art** in PC 10.1; may include organizational chart, pyramid, Venn diagrams, etc.
- **Custom animation effects** in PC 13.1; may include adjusting the animations assign in the master slide

Critical aspects

The assessment must confirm that the candidate is;

- Ability to perform all presentation functions included in this unit.

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Multimedia capable computer • Printer • Scanner • Micro phone	• Paper • Printing cartridge for printer

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Intellectual Property Act No. 36 of 2003
- Shelly Cashman Series MicrosoftOffice 365 & Office 2019 Introductory (MindTap Course List) 1st Edition, by Sandra Cable (Author), Steven M. Freund (Author), Ellen Monk (Author), Susan L. Sebok (Author), Joy L. Starks (Author), ISBN : 0357026438, 978-0357026434, Cengage Learning, 2019

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• English literacy. • Basic knowledge of computer configuration and operations, main components of computers, their functions and mode of operations. • Standard operating systems. • Basic knowledge of computer viruses and spy ware and common methods of protection. • F file management. • Effective presentation guidelines.	• Ability to connect relevant external parts and peripherals of a computer. • Ability to power on and boot up a computer • Ability to perform basic operations on a computer, manage files etc.

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Perform basic operations of cloud computing including internet and email services
Unit Descriptor:	This unit covers the competencies required to Access and use Internet services available, send, receive, handle emails, sign in to cloud services, access basic services provided by cloud servers, access and use social media sites safely.
Unit Code:	K72S004U05
Element	Performance Criteria
1. Access Internet services	1.1 Required Web browser selected and opened 1.2 Navigation buttons used as required 1.3 Refresh/Reload, stop buttons used as appropriate 1.4 Bookmarks/favorites managed as required 1.5 Web pages saved for offline access as and when required 1.6 Browser settings such as changing default home page customized 1.7 Search engine/s selected as required 1.8 AI based content search engine/s selected as appropriate 1.9 Required files such as documents, audios, videos, images, software searched and downloaded 1.10 Temporary Internet files and search history cleared 1.11 plug-in/add-on/extensions added/removed and setup as required 1.12 Whole or part of a web page printed as required
2. Use email services	2.1 Email account created / signed up as required as per the standard procedure 2.2 Created email account signed in 2.3 Email checked as required 2.4 Email sent using cc/Bcc as required 2.5 Email sent with attachment/s using required option
3. Work with social media/sites	3.1. Social media sites accessed following the best practices 3.2. Contents added to the social networks/medias as required 3.3. YouTube channel created and managed as required
4. Use software applications collaboratively	4.1 Word processing files created/shared for collaborative edits with <i>different permission settings</i> 4.2 Spread sheet files created/shared for collaborative edits with different permission settings 4.3 Presentations created/shared for collaborative edits with different permission settings 4.4 Data/information collection forms created and used for data collections purposes

5. Use software applications for video conference	5.1 Scheduled video conferencing created with different <i>participations and host permission privileges</i> 5.2 Video conferencing hosted with <i>different practical situations</i> and adhering to general ethics/organizational procedures 5.3 Attendance of the participants kept if available 5.4 Recordings of the meetings kept if available
6. Handle cloud storages	6.1 Files and documents uploaded to the cloud storage as per the SOP 6.2 Backup and restore facilities used as per requirements 6.3 Cloud spaces/folders shared with necessary parties as per the requirements 6.4 Cloud storage/local folders automatic syncing setup as per the requirements 6.5 Cloud storage accessed as required

Range Statement:

The work outlined in this unit shall be demonstrated individually in an office environment or in a computer laboratory and shall include all the competencies required to use electronic mail and the internet efficiently and safely.

Elaboration of Terms in Performance Criteria

- **different permission settings** in PC 4.1: View only, editing provision, download restrict, comment only etc.
- **participations and host permission privileges** in PC 5.1: Mic/camera permissions, Screen share permissions, recording permissions, participation restrictions etc.
- **different practical situations** in PC 5.2: Share screen, shared application window only, host an online meeting with a multimedia projector etc.

Critical aspects

The assessment must confirm that the candidate is able to;

- Search and download required documents. Images, Audios and Video efficiently
- Send/forward emails with attachments and with CC/ BCC options as required
- Handle a video conferencing session accurately
- Upload and download files in the cloud storage

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Computer with internet facilities • Multimedia projector • Web cam • Microphone/ headphone	• Stationaries

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Intellectual Property Act No. 36 of 2003
- Microsoft Office 365 for Beginners 2023: [8 in 1] , EDWARD P. DERRICK, 979-8363628559, Independently published, 2022

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Setup computer peripherals • Time zones for global meetings • Cultural difference and greetings for global meetings	• Language skills • Communication skills • Time management • Punctuality • Adhering to general ethics • Coordinating skills

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Handle relational database
Unit Descriptor:	This unit covers the competencies required to create and manipulate a database as required using a standard interface
Unit Code:	K72S004U06
Element	Performance Criteria
1. Perform basic operations	1.1 A database application opened and closed properly. 1.2 A new blank database created as required. 1.3 A new database saved to a specified location on a drive. 1.4 Database closed 1.5 A required database opened.
2. Create and modify tables	2.1 Table object selected 2.2 Table created with appropriate field names, data types and attributes. 2.3 Validation rules applied for fields as required 2.4 Fields inserted/ modified /deleted as required.
3. Define keys and built relationships	3.1. A primary key defined as per the standard procedure. 3.2. A field indexed with / without duplicates using required option. 3.3. Relationships built as per the standard
4. Manipulate Table data using datasheet / form	4.1. Records added, modified, deleted using datasheet view as required. 4.2. Forms created using single table / multiple tables as required 4.3. Table structure/ table with data/ only data imported as required 4.4. Data exported as required
5. Create and modify Queries	5.1 Query created using a single table / multiple tables with required criteria 5.2 Aggregated function/s added to the query as required. 5.3 A query edited by adding, modifying or removing criteria as required.
6. Create and modify reports	6.1 A report based on a table, query created and saved as required. 6.2 The arrangement of data fields and headings within a report layout changed as required. 6.3 Data grouped and sorted in the report as required. 6.4 Aggregated function/s added to the report as required. 6.5 Header, footer added or modified as necessary. 6.6 A report saved and closed as per the correct procedure.
7. Print required information	7.1 A table structure, form layouts, report layouts previewed and printed when necessary 7.2 Query results previewed and printed as required 7.3 Reports generated, previewed and printed as required

Range Statement:

The work outlined in this unit shall be demonstrated individually in an office environment or in a training center computer laboratory and shall include all the competencies required to create, update and maintain a database in a computer, and extract data as required. The database can exist maximum of three tables with the appropriate relationships.

Elaboration of Terms in Performance Criteria

- **Required criteria** in PC 5.1 may include comparison operators such as < (Less than), <= (Less than or equal to), > (greater than), >= (greater than or equal to), = (equal), <> (not equal to) and logical operations such as “OR” and “AND”.
- **Aggregated function/s** in PC 6.4 may include SUM (), AVERAGE (), MIN (), MAX () and COUNT ()

Critical aspects

The assessment must confirm that the candidate is should be able to;

- Define primary keys as appropriate
- Build the relationship as appropriate
- Create forms
- Create queries
- Create reports

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Computer with standard operating system and database application. • Printer	• Paper / print media • Printing cartridge for printer

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Intellectual Property Act No. 36 of 2003
- Access For Dummies, Laurie A. Ulrich, Ken Cook, John Wiley & Sons, 2021, ISBN 1119829089, 9781119829089

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• English literacy. • Basic computer configuration and operations, main components of computers, their functions and mode of operations. • Standard operating systems. • Computer viruses and spyware and common methods of protection. • File management.	• Ability to connect relevant external parts and peripherals of a computer. • Ability to power on and started up a computer • Handling a printer • Use of help function

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Conduct routine maintenance services of computer system and peripherals
Unit Descriptor:	This unit covers the competencies required to identify the basic structure of a computer system, peripheral devices and faults. Install, configure, update, uninstall operating system and application software. Identify network devices, configuration and troubleshoot on computer networks.
Unit Code:	K72S004U07
Element	Performance Criteria
1. Identify and Setup computer & network devices	1.1 Computer peripherals, network devices and power management devices identified and functionality ensured 1.2 Peripherals and network devices installed and configured as per the manufacturer's guidelines
2. Rectify faults	2.1 Operating system installed/reinstalled/ uninstalled/configured 2.2 Application software installed/reinstalled/uninstalled/configured 2.3 Operating system/Application software updated as available and required 2.4 Device drivers installed and updated as necessary 2.5 Hardware and network issues identified and corrected / escalated as per the SOP
3. Maintain computer system	3.1 Antivirus software installed, configured and updated 3.2 Virus scan and clean performed as necessary 3.3 Disk cleanup and defragmentation performed as and when required 3.4 Backups and restore performed as per the SOP 3.5 Problems reported to authorized officer as per the SOP 3.6 Records of maintenance requests/related issues maintained as per the SOP

Range Statement:

Work may take place in this unit shall be demonstrated individually in an office environment or in a computer training center laboratory with LAN (both wired and wireless)

Elaboration of Terms in Performance Criteria

- **Peripherals** in PC 1.1 & 1.2 may include printers, network printers, Scanners, multimedia projectors etc
- **Network devices** in PC 1.1 & 1.2 may include network switch, Routers (ADSL / Fiber/wireless), Wireless access point etc.
- **Power management devices** in PC 1.1 may include UPS

Critical aspects

The assessment must confirm that the candidate is able to;

- Identify the configuration of hardware / software /network
- Identify the hardware / software /network faults
- Setup hotspot connectivity

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Computer • Printer • Scanner • Router (ADSL / Fiber/wireless) • Storage Devices • LAN Cable tester • Wireless access point • Network switch • Network diagnostic tools • Operating system software • Antivirus software • Crimping tool	• Network cable • Wall outlets • RJ 45 Connectors

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Payment Devices Frauds Act No. 30 of 2006.
- Intellectual Property Act No. 36 of 2003

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Computer and peripheral devices • Basic knowledge of computer security • Basic Networking principles. • Operating systems Principles • Knowledge of safety (Personal & equipment) • Cabling standards • Basic knowledge of Wireless technology	• Communication skills • Following safety procedures • Cabling skills • Read and interpret technical documents

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Produce Print Page Layout and multimedia objects
Unit Descriptor:	This unit covers competencies required to produce Print Page Layout & multimedia objects.
Unit Code:	K72S004U08
Element	Performance Criteria
1. Develop a basic design concept for Page layout	1.1. Customer requirement identified 1.2. Basic design requirements (Purpose/endues, audience, budget, Deadline) identified 1.3. Tentative Thumbnail sketches produced 1.4. Customer's approval obtained
2. Gather graphics & Illustration for Page layout	2.1 Images, logos and illustrations gathered as required for the design and suitability for reproduction ensured. 2.2 Suitable scanner to scan images and illustrations selected and scanned according to required resolutions and size Illustrations/graphics created as required 2.3 Images/graphics saved in a separate folder in appropriate file format according to organizational guidelines
3. Design Page Layout	3.1 File Medium format prepared according to customer requirement and the agreed design concept. 3.2 Layout Template produced according to agreed design concept. 3.3 Text formatting performed according to agreed design solution 3.4 Proofreading performed to ensure accuracy of the final product 3.5 Vector and Raster graphics requirements identified for the agreed design solution. 3.6 Perform Final proof submission and obtaining of client's approval according to industry/company procedure
4. Design multimedia products for social media / web sites	4.1 Client's requirement identified. 4.2 Requirement gathered of the multimedia described 4.3 Relevant multimedia tools of mobile & computer base applications and formats /web tools identified 4.4 Multimedia products for social media/websites edited as per the client's requirement and obtained approval. 4.5 Multimedia source file saved as per the organizational guidelines 4.6 Final product uploaded/delivered as per the client as required

Range Statement:

Work may take place in a small office environment or computer laboratory

Elaboration of Terms in Performance Criteria

- **Thumbnail sketches** in PC 1.3 may include the miniature Hand drawn image of the design
- **File Medium Format** in PC 3.1 may include the size of the page, Number of pages, binding method, finishing method, etc.

Critical aspects

The assessment must confirm that the candidate is;

- Develop design concept, layout, images and other graphic objects using graphic design software
- Followed due approval process
- Develop web page as per the client requirement
- Identify animated motion to the developed graphic

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Graphic design software • Vector/Raster / motion graphics and video editing software • PC with sound system & standard operating system • SMART phone • Printer • Scanner • Headphone	• Stationary

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Intellectual Property Act No. 36 of 2003
- The Ultimate Guide to Graphic Design - A MacUser Publication Edited by Nik Rawlinson
- The Non-Designer's Design Book - Robin Williams
- The Fundamentals of Graphic Design Gavin Ambrose and Paul Harris

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Fundamentals of graphic design • Computer hardware operating system and application software • File types and file compression techniques • Storage types	• Basic computer skills • Skills in handling scanner, printer etc. • Communication skills • Esthetic and creative skills

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Develop information system
Unit Descriptor:	This unit covers the competencies in gathering and analyzing requirements, design, develop, test, implement, document & maintain basic information system
Unit Code:	K72S005U09
Element	Performance Criteria
1. Gather requirements for information system	1.1 Data gathered using standard <i>data gathering techniques</i> according to the customer's requirements- 1.2 <i>Feasibility study</i> conducted as per the SOP 1.3 Software Requirement Specification (SRS) documented as per organizational standards 1.4 SRS reviewed and final approval obtained
2. Design information system	2.1 Data Flow Diagrams (DFD) developed according to the system requirement 2.2 Data Flow Diagrams reviewed and finalised 2.3 Flowcharts developed according to the system requirement 2.4 Pseudocodes developed according to the developed flowcharts 2.5 Required UML (Unified Modelling Language) diagrams developed 2.6 Test plan prepared as per system requirement
3. Implement information system	3.1. Software developed using the selected programming language according to the developed specification 3.2. Databases connected as per the requirement 3.3. Developed software tested as per test plan 3.4. <i>System specification documentation</i> prepared as per the SOP

Range Statement:

Work may take place in a small-scale in-house development environment or training center with computer facility Work may include analyze, design and develop information system for individual or small and medium scale usage diagrams may include use case diagram and class diagram

Elaboration of Terms in Performance Criteria

- ***data gathering techniques*** in PC 1.1 may include Document/Report review, Interview, questionnaires and observation
- ***Feasibility study*** in PC 1.2 may include Technical, financial, practical and legal feasibility
- ***System specification documentation*** in PC 3.4 may include User guide, DFD, Flow charts, ER diagram, UML diagrams, Test cases

Critical aspects

The assessment must confirm that the candidate is able to;

- Draw a level 2 DFD
- Draw a suitable flow chart for given scenario
- Design a user interface and make connection to the relevant database

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Computer with standard operating system • Software development packages and tools • Diagrammatic tools • Printer	• stationaries

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Intellectual Property Act No. 36 of 2003
- Modern Systems Analysis and Design, Ninth Edition , Joseph S. Valacich, Joey F. George, 2021, **Pearson Education**

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Basic IT • Fundamentals of programming • Object oriented concepts • System development methodologies • Basic database concepts	• Basic computer Skills • Communication Skills • Programming language skills • Documentation skills

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Handle relational databases using standard Structured Query Language (SQL)
Unit Descriptor:	This unit covers the competencies required to create and manipulate a database as required using standard SQL statements.
Unit Code:	K72S004U10
Element	Performance Criteria
1. Startup Database Management System (DBMS)	1.1. Selected DBMS Installed and setup 1.2. User logged in to DBMS as per the SOP
2. Manipulate Database & Tables	2.1 Database created as required 2.2 Tables created using relevant column names, data types and field sizes with/without primary keys as required 2.3 Columns inserted, deleted and altered as required. 2.4 Foreign keys defined as required when creating the tables 2.5 Primary keys and foreign keys Added, modified & deleted as required 2.6 Table structures imported/exported as required 2.7 Backup/Restore database as required 2.8 Databases /Tables deleted as per the SOP
3. Handle data	3.1 Records inserted to the table/s as required 3.2 Records modified in the table/s as required 3.3 Records deleted in the table/s as required 3.4 Data imported/exported as required
4. Create Queries	4.1 Query created to retrieve data from single table with/without conditions 4.2 Query created to retrieve data from multiple tables with/without conditions 4.3 Aggregated function/s added to the query as required.
5. Manipulate views	5.1 Views created as required 5.2 Views modified as required 5.3 Views deleted as required

Range Statement:

The work outlined in this unit shall be demonstrated individually in an office environment or in a computer laboratory and shall include all the competencies required to create, update and maintain a database in a computer, and extract data as required using standard SQL statements. The database can exist maximum of three tables with the appropriate relationships.

Elaboration of Terms in Performance Criteria

- **Aggregated function/s** in PC 4.3; may include SUM (), AVERAGE (), MIN (), MAX () and COUNT ()

Critical aspects

The assessment must confirm that the candidate is;

- Create tables with relevant primary keys
- Define foreign keys as appropriate
- Create queries

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• Computer with standard operating system and data base application. • SQL Server Management Studio/ MySQL /Maria DB/WAMP / XAMPP	• Paper / print media • Printing cartridge for printer

The following documents /References/Standards may be used for this unit:

- Sams Teach Yourself SQL in 24 Hours, SIXTH EDITION, Ryan Stephens, Arie D. Jones, Ron Plew 2016 by Pearson Education ISBN: 978-0-672-33759-8, 0-672-33759-2

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• English literacy. • Basic knowledge of computer configuration and operations, main components of computers, their functions and mode of operations. • Standard operating systems. • Basic knowledge of computer viruses and spyware and common methods of protection. • Knowledge of file management. • Basic knowledge of databases.	• Ability to connect relevant external parts and peripherals of a computer. • Ability to power on and started up a computer • Create a new database, open a saved database, enter and manipulate data and format as required. • Generate reports as required from the information contained in the database. • Take printouts of part or whole of a report.

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Design static Website
Unit Descriptor:	This unit covers competencies required to create static Web Pages using Hypertext Markup Language (HTML) and Cascading Style Sheets (CSS)
Unit Code:	K72S004U11
Element	Performance Criteria
1. Identify customer requirement	1.1 Layout of the website design according to the customer requirement 1.2 Design layout finalized as per the organizational procedure 1.3 Web designing best practices identified
2. Create a web page using HTML	2.1 Basic structure of the web page identified 2.2 HTML elements identified 2.3 Web pages created using HTML tags and attributes 2.4 Objects inserted on the webpage 2.5 Web pages linked using hyperlinks
3 Enhance the view of the web pages using CSS	3.1 Basic CSS syntax identified 3.2 CSS applying methods identified 3.3 CSS codes applied as required

Range Statement:

Work may take place in a small office environment or computer laboratory

Elaboration of Terms in Performance Criteria

- **Objects** in PC 2.4 may include tables, images, audio, video and animations

Critical aspects

The assessment must confirm that the candidate is;

- Develop a web site with minimum 3 pages using HTML and CSS as per the client requirement

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• PC with standard operating system • Printer	• HTML editing software

The following documents /References/Standards may be used for this unit:

- Computer Crimes Act No. 24 of 2007.
- Payment Devices Frauds Act No. 30 of 2006.
- Intellectual Property Act No. 36 of 2003
- Basics of Web Design HTML5 & CSS Fifth Edition, Terry Ann Felke-Morris, 2020 by Pearson Education
- Beginners Guide to Web Design, S.B. Chatto, 2020, ISBN: 978-978-968-419-9

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Computer hardware operating system and application software • Basic computer application literacy • Basic understanding of World Wide Web (WWW)	• Basic computer skills • Skills in handling scanner, printer etc. • Communication skills • Esthetic and creative skills

Worker behavior/Attitudes

Attitudes/Behavior	
• Responsibility • Self esteem • Self confidence • Sociability skills • Self-management • Integrity and honesty	• Punctuality and efficiency • Cooperative • Self-motivated

Unit Title:	Practice workplace communication and interpersonal relations
Unit Descriptor:	This unit covers the competencies required to co-operate with others, interact effectively and deal with information within the workplace.
Unit Code:	K72S004BU01
Element	Performance Criteria
1.Co-operate with others in performing work	1.5 Work performed in co-operation with others in a friendly & courteous manner 1.6 Verbal and non-verbal communication used appropriately 1.7 Ideas, opinions and interactions shared & respected appropriately 1.8 Work performed in co-operation with co-workers and related technical staff appropriately
2.Work with people of diverse backgrounds	2.1 People of a variety of backgrounds accepted without prejudice 2.2 Respect for cultural, ethnic and gender differences practised appropriately
3. Obtain record and convey workplace information	3.1 Effective questioning, active listening reading and speaking skills used to gather information 3.2 Information and ideas transferred without distortion 3.3 Work place interaction conducted in a courteous manner 3.4 Approved work instructions obtained and followed in appropriate manner 3.5 Collected information recorded for easy reference
4. Fill work related documents	4.1 Work related forms and formats identified appropriately 4.2 Range of forms and formats relating to work filled, appropriately 4.3 Reporting requirements to supervisor completed accurately in accordance with organizational guidelines and timelines.

Range statement

The performance of activities included in this unit may take place in workplace. Communication will include verbal, non-verbal, written, printed and electronic forms. It applies to individuals, groups and co-workers as well as others.

Critical Aspects:

The assessment must confirm that the candidate is able to;

- Obtaining accurate information from customer/superior/co-workers
- Accurate recording of information (Defects)
- Conveying information to relevant parties accurately

The following tools, equipment & material are included within this unit.

Tools/ instruments and equipment

- Pens/pencils/rulers/calculator
- Paper/notebooks
- Telephones

Materials

- Charts
- Graphics
- Memos
- Formats & Forms (Time Record sheets, Quality record sheets and material order sheets)

The following documents /References/Standards may be used for this unit:

- Standard Organizational Procedures and accepted ethical standards and norms

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Safety symbols & procedures • Mother tongue in both spoken and written form • Basic English relevant to workplace and type of work • Organisation's rules, regulations and procedures • General norms of polite conduct • Forms of address appropriate to customers and persons in different positions • Basic knowledge of customs and practices of different ethnic and religious groups • Good telephone manners • Basic understanding of gender issues • Basic knowledge of new trends in ICT sector	• Speak clearly in mother tongue and express varied ideas • Write clearly and legibly in mother tongue • Read and understand simple English relevant to the workplace • Write simple explanations in English language • Communicate effectively using the telephone • Perform simple calculations • Make simple sketches

Unit Title:	Apply occupational literacy and numeracy
Unit Descriptor:	This unit covers the competencies required to apply literacy and Numeracy to participate effectively in activities in the workplace.
Unit Code:	K72S004BU02
Element	Performance Criteria
1. Employ basic mathematics to complete work tasks and solve problems	1.1 Computation skills demonstrated using whole numbers, fractions, decimals and percentages manually and with calculators. 1.2 Key concepts and principles of mathematics applied to understand and solve workplace calculations. 1.3 Key concepts and principles of basic geometry applied to understand their use in the workplace. 1.4 Computation skills demonstrated using ratios
2. Communicate ideas in writing	2.1 Adequate vocabulary maintained for everyday tasks according to nature of work 2.2 Written material produced to convey intended message in familiar work context 2.3 Appropriate units, codes and symbols selected according to purpose of written material
3. Read range of written materials	3.1 Adequate vocabulary maintained to read range of written materials according to workplace requirements 3.2 Appropriate units, codes and symbols interpreted for the purpose of reading 3.3 Information gained from text integrated with own knowledge in order to create meaning

Range/Context:

Activities included in this unit may take place in workplace, and will include but not be limited to reading reports, instruction sheets, technical manuals and other types of written material in the mother tongue as well as in simple English, writing in the mother tongue and in simple English sentences, and performing simple work-related calculations employing basic mathematical operations.

Critical Aspects:

The assessment must confirm that the candidate is able to;

- Interpret written material accurately
- Convey ideas clearly and correctly in writing
- Adopt appropriate method in mathematical calculations

The following tools, equipment & material are included within this unit.

Tools/ instruments and equipment

- Pens/pencils
- Paper/notebooks
- Calculator

Materials

- Charts
- Manuals
- Memos
- Formats & Forms (Time Record sheets, Quality record sheets and material order sheets)
- Accepted rules of grammar and syntax
- Fundamental principles of mathematics

The following documents /References/Standards may be used for this unit:

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Underpinning Knowledge, Skills

Underpinning Knowledge	Underpinning Skills
• Grammar, spelling and syntax in mother tongue • Basic English grammar, spelling and syntax • Mathematical representation of physical quantities • Basic Mathematical operations • Adequate technical vocabulary in English • Principles of basic of Geometry	• Write clearly, legibly and correctly in mother tongue • Read and understand simple English relevant to the workplace • Write simple sentences in English • Perform simple calculations • Draw simple Geometrical views

Unit Title:	Work in teams
Unit Descriptor:	This unit covers the competencies required to work collaboratively and in cooperation with others, and identify roles & responsibilities of members of the team.
Unit Code:	K72S004BU03
Element	Performance Criteria
1.Understand team role and scope	1.1 Formal and informal teams and their roles and limitations identified in workplace 1.2 Responsibilities, characteristics and relationship of team members explained to accomplish team goals
2.Participate as a team member	2.1 Different forms of communication used with team members to contribute to team objectives 2.2 Appropriate contribution (participation) made to the team to meet team goals 2.3 Safety of team members ensured when performed work operations appropriately 2.4 Different viewpoints and ideas respected as required to meet team goals 2.5 Collective decisions accepted as required to meet team goals
3. Maintain productive group relations	3.1 Individual's role and responsibility within the team clearly identified 3.2 Other's roles and responsibilities identified and understood 3.3 Harmonious relations maintained and respected within the team and with others

Range/Context:

The performance of activities included in this unit may take place in workplace and will include but not be limited to identifying individual's and other's roles and responsibilities within the team and maintaining harmonious relations.

Critical Aspects:

The assessment must confirm that the candidate is able to;

- Achieve team goals and individual goals
- Follow designated work plan for the job
- Respect to other's ideas

The following tools, equipment & material are included within this unit.

Tools/ instruments and equipment

- Depending on the job assigned to the team for the purpose of assessing this unit, required tools, equipment and material will be provided.

Materials

- Work plans
- Manuals
- Handbooks
- General rules and norms guiding team behavior

Underpinning Knowledge and Skills**Underpinning Knowledge**

- Simple concepts of team behaviour
- Elementary group dynamics
- Basic knowledge in social and demographic structures

Underpinning Skills

- Work effectively and comfortably as a member of a team
- Patient
- Understand others

Unit Title :	Practice occupational health and safety procedures in a workplace
Unit Descriptor :	This unit describes the performance outcomes required to identify and follow safety and emergency procedures in workplace. It involves those safety procedures to be followed when using workplace hand tools and hand-held power tools, fixed equipment, and chemicals It includes the individual's responsibility for safety and emergency response in the event of an accident or incident.
Unit Code:	K72S004BU04
Element	Performance Criteria
1. Identify and follow workplace safety procedures	1.1 Work place procedure relating to personal safety identified and followed 1.2 Unsafe situations and hazards/risks identified and reported according to workplace 1.3 Machinery and equipment breakdowns identified and reported according to workshop procedures 1.4 Fire hazards are identified, different fire protection equipment and material and their methods of use identified as required for different types of fires 1.5 Work place procedures and safety requirements for handling and storing dangerous goods and hazardous substances identified and followed 1.6 Manual handling procedures and ergonomic parameters identified and followed 1.7 Health and Safety awareness program participated 1.8 Appropriate safety gear used/ worn according to organisation's requirements & manufacturer's instructions 1.9 First aid procedures performed as needed 1.10 Daily and weekly safety & health check list followed as per organization requirement
2. Identify and follow workplace emergency procedures	2.1 Mechanical, electrical, chemical and biological hazards recognized, identified and reported promptly. 2.2 Workplace emergency procedures regarding illness and accidents are identified and followed 2.3 Safety warning alarms and workplace evacuation procedures, emergency protection area and fire exits identified and

	followed
2.4	Firefighting appliances and equipment located and identified for emergency use and application
2.5	Relevant persons identified for contacting in the event of an incident, accident or sickness of self, co-workers, staff, customers and others
2.6	Accident and incident documentation requirements followed work place procedures

Range/Context:

Activities included in this unit may take place in a work place, and will include but not be limited to identifying potential hazards, taking effective preventive action to control or minimise such hazards, using personal protective equipment, using fire extinguishers and maintaining records of safety related activities.

Elaboration of terms:

Workplace procedures must include procedures for:

- Selecting and using personal protective equipment (PPE)
- Identifying hazards
- Emergency, fire and accident
- Regular cleaning of uniforms

First aid procedures may include but not be limited to treatment of minor cuts, bruises and burns, applying bandages and tourniquets, and neutralising the effects of corrosive substances.

Relevant persons must include:

Those holding a current recognized qualification in any of the following:

- First aid
- Cardiopulmonary resuscitation
- Fire warden and workplace evacuation
- Occupational or work health and safety representative.

Critical Aspects:

- Identify potential hazards
- Adhere to safety procedures
- Use most appropriate tools for a task

The following tools, equipment & material are included within this unit.

Tools/ instruments and equipment

- Safety signs and symbols
- Posters
- Coloured barricaded tape and paint
- Appropriate machine guards
- Machine and safety devices
- Safety shoes
- Goggles
- Gloves
- Ear protectors
- Safety belts
- Helmets
- First aid box with essential items
- Fire protection equipment

Materials

-

The following documents /References/Standards may be used for this unit:

- Health & safety procedure manual
- Organisation's health & safety procedure manual
- Health & safety procedure manual
- Factories' Ordinance No.45 of 1942 and its amendments

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
Individuals must be able to demonstrate knowledge of: basic aspects of relevant OHS legislation, including: • Employer responsibilities for workplace • Employee responsibilities to participate in health and safety practices • Employee responsibility to ensure own work practices protect the safety of self, other workers and other people in the workplace roles of work place personal safety in workshop, including: • Hearing protection • Skin protection • Protective clothing • Protective footwear • Eye protection manual handling, including:	• Learning skills to: Locate sources of information relating to safety requirements. • Reading skills to: Interpret information from manufacturer and literature when seeking procedures required carrying out work safely. • Writing skills to: Legibly and accurately enter information into work place safety documents and check sheets. • Oral communication skills to: Clarify instructions inform personnel of work place hazards. • Numeracy skills to: Match hazardous material weights and volumes to safety data sheets (SDS) and work instructions. • Planning and organizing skills to: Plan own work requirements to achieve required outcomes while working according

- Storing items
 - Planning the lift
 - Using correct lifting techniques and equipment
- work place practices, including:
- Housekeeping
 - Observing personal hygiene
 - Identifying slip hazards and trip hazards
- OHS symbols and signs, Machine and equipment safety, including:
- Using machine guards
 - Electrical tools, including:
 - Checking before use faulty equipment tagging and isolation procedures
 - Dangers of using electrical tools around water
 - Electrical Hazards
- emergency procedures, including:
- Personnel responsibilities
 - Evacuation procedures and assembly points.
- to safety requirements.
- Self-management skills to:
 - Follow work place safety requirements without constant supervision identify and look after own safety.
 - Problem solving skills to:
 - recognize when unsure of safety procedures and seek help
 - Recognize potential workplace hazards and take action.

**COMPETENCY BASED TRAINING CURRICULUM
FOR
INFORMATION AND COMMUNICATION TECHNOLOGY
TECHNICIAN
NVQ Level 2, 3 and 4**

List Of Curriculum Modules		Module Code No.	Duration		Page No
			Theory	Practical	
1.	Maintaining files & folders	K72S004M01	18	06	47
2.	Performing word processing	K72S004M02	24	36	48-49
3.	Preparing spread sheets	K72S004M03	24	36	50-51
4.	Preparing presentations	K72S004M04	12	18	52
5.	Performing basic operations of cloud computing including internet and email services	K72S004M05	24	24	53-54
6.	Performing basic database manipulation	K72S004M06	24	36	55
7.	Computer system, network and peripherals	K72S004M07	36	24	56-57
8.	Producing Print Page Layout and multimedia objects	K72S004M08	30	48	58
9.	Systems Analysis and Designing	K72S004M09	36	12	59-60
10.	Perform relational databases using standard SQL	K72S004M10	24	36	61-62
11.	Designing static Website	K72S004M11	12	18	63
12.	Developing Desktop applications	K72S004M12	48	36	64-65
13.	Communication skills for workplace	K72S004BM01	24	00	66
14.	Occupational literacy and numeracy	K72S004BM02	30	00	67
15.	Team Work	K72S004BM03	06	00	68
16.	Occupational Safety & Health and Environmental Aspects	K72S004BM04	18	00	69-70

Competency Profile

FLIGHT DISPATCHER

COMPETENCY AREAS

COMPETENCIES

A. Maintain files & folders	A1. Start up and turn off the computer	A2. Customize computer and desktop settings	A3. Create folders and files	A4. Perform Folder/File operations	A5. Set attributes of files folders
	A6. Compress and extract folder/ files	A7. Backup and restore folders/ files			
B. Perform word processing	B1. Create different types of documents using templates	B2. Document Save, Close and Open	B3. Adjust word processor setting	B4. Make text processing operations	B5. Format text, paragraph and document
	B6. Create & format tables	B7. Add objects and special formatting	B8. Create links	B9. Create mail merge	B10. Find & replace text
	B11. Create templates/cover page	B12. Draw graphics/auto text	B13. Create Table of Content/Table of figures	B14. Print a word document/envelopes/label	
C. Prepare spread sheet	C1. Format work sheet	C2. Apply built – in functions and formulas	C3. Fill, Sort and Filter data	C4. Prepare graphs and charts	C5. Use data tools
	C6. Create a pivot table	C7. Develop and customize worksheet	C8. Set Page properties and print a spread sheet		
D. Prepare presentations	D1. Creating a presentation	D2. Make basic operations with presentation software	D3. Adjust settings	D4. Change views	D5. Add/Delete /Move slides
	D6. Apply design templates to an existing presentation	D7. Apply slide master	D8. Insert different media	D9. Slide transition and animation	D10. Insert action buttons and hyperlinks
	D11. Conduct the presentation	D12. Print slides			
E. Perform basic operations of cloud computing including internet and Email services	E1. Perform Browser operations	E2. Use Search engines	E3. Download and upload file	E4. Perform e mail operations	E5. Work with social media/sites
	E6. Apply software applications collaboratively	E7. Apply software applications for video conference	E8. Handle cloud storages		
F. Maintain relational databases	F1. Create tables	F2. Create table relationship	F3. Modify tables	F4. Create forms	F5. Create queries
	F6. generate reports	F7. Modify records	F8. Backup and restore data	F9. Link data base with external document	

G. Conduct routine maintenance services of computer system and peripherals	G1. Identify computer & network devices	G2. Check hardware compatibility with operating system	G3. Prepare the computer for installation	G4. Install the operating system	G5. Install device drivers
	G6. Identify basic hardware & network issues	G7. Correct basic hardware & network issues	G8. Detect operating system errors	G9. Install application software	G10. Install antivirus software
	G11. Ensure standard performance	G12. Record maintenance requests/related issues			
H. Produce Print Page Layout and multimedia objects	H1. Identify customer requirement	H2. Identify basic design requirement	H3. Form thumbnail sketches	H4. Collect Images, logos and illustrations as required	H5. Save images/graphics
	H6. Set file medium format	H7. Form layout template	H8. Make text formatting according to agreed design solution	H9. Make proof reading	H10. Identify Vector and Raster graphics requirement
	H11. Ensure accuracy of the final product	H12. Ensure accuracy of the final product	H13. Make final proof submission with client's approval		
I. Develop basic software solution	I1. Identify software solution requirement	I2. Prepare system diagrams and algorithm	I3. Develop coding and connect database	I4. Test developed software	I5. Package the developed software
	I6. Prepare System specification documentation	I7. Implement the developed software			
J. Handle relational databases using standard SQL	J1. Install and Setup the selected DBMS	J2. Create Database	J3. Create tables	J4. Create table relationship	J5. Modify tables
	J6. Handle data	J7. Backup and restore data	J8. Import /Export data	J9. Create queries	J10. Deploy views
K. Design static web pages	K1. Identify customer requirement	K2. Identify basic structure of the web page	K3. Identify HTML elements	K4. Use HTML tags and attributes	K5. Insert object on the webpage
	K6. Link webpages using hyperlinks	K7. Identify basic CSS syntax	K8. Apply CSS methods	K9. Apply CSS codes	

Programme Structure

Competency Area		Module		Competencies	Time	
		No.	Title		Institutional	
					Theory	Practical
A	Maintain files & folders	M 01	Maintaining files & folders	A01, A02, A03, A04, A05, A06, A07	12	06
B	Perform word processing	M 02	Performing word processing	B01, B02, B03, B04, B05, B06, B07, B08, B09, B10, B11, B12, B13, B14	24	36
C	Prepare spread sheet	M 03	Preparing spread sheets	C01, C02, C03, C04, C05, C06, C07, C08	24	36
D	Prepare presentations	M 04	Preparing presentations	D01, D02, D03, D04, D05, D06, D07, D08, D09, D10, D11, D12	12	18
E	Perform basic operations of cloud computing including internet and email services	M 05	Performing basic operations of cloud computing including internet and email services	E01, E02, E03, E04, E05, E06, E07, E08	24	24
F	Performing basic database manipulation	M 06	Performing basic database manipulation	F01, F02, F03, F04, F05, F06, F07, F08, F09	24	36
G	Conduct routine maintenance services of computer system and peripherals	M 07	Computer system, network and peripherals	G01, G02, G03, G04, G05, G06, G07, G08, G09, G10, G11, G12	36	24
H	Produce Print Page Layout and multimedia objects	M 08	Producing Print Page Layout and multimedia objects	H01, H02, H03, H04, H05, H06, H07, H08, H09, H10, H11, H12, H13	30	48

I	Analyze, design and develop information system	M 09	Systems analysis and design	I01, I02	36	12
		M12	Developing Desktop applications	I03, I04, I05, I06, I07	48	36
J	Handle relational databases using standard SQL	M 10	Handling relational databases using standard SQL	J01, J02, J03, J04, J05, J06, J07, J08, J09	24	36
K	Design static Website	M11	Designing static Website	K01, K02, K03, K04, K05, K06, K07, K08, K09	12	18

Mapping Table

Qualifications	Relevant Units	Relevant Modules	Relevant Tasks
K72S004Q1L2	U01	M01	A01, A02, A03, A04, A05, A06, A07
	U02	M02	B01, B02, B03, B04, B05, B06, B07, B08, B09, B10, B11, B12, B13, B14
	U05	M05	E01, E02, E03, E04, E05, E06, E07, E08
	BU1, BU2, BU3, BU4	BM1, BM2, BM3, BM4	
K72S004Q2L3	U01	M01	A01, A02, A03, A04, A05, A06, A07
	U02	M02	B01, B02, B03, B04, B05, B06, B07, B08, B09, B10, B11, B12, B13, B14
	U03	M03	C01, C02, C03, C04, C05, C06, C07, C08
	U04	M04	D01, D02, D03, D04, D05, D06, D07, D08, D09, D10, D11, D12
	U05	M05	E01, E02, E03, E04, E05, E06, E07, E08
	U06	M06	F01, F02, F03, F04, F05, F06, F07, F08, F09
	U07	M07	G01, G02, G03, G04, G05, G06, G07, G08, G09, G10, G11, G12
	BU1, BU2, BU3, BU4	BM1, BM2, BM3, BM4	
K72S004Q3L4	U01	M01	A01, A02, A03, A04, A05, A06, A07
	U02	M02	B01, B02, B03, B04, B05, B06, B07, B08, B09, B10, B11, B12, B13, B14
	U03	M03	C01, C02, C03, C04, C05, C06, C07, C08
	U04	M04	D01, D02, D03, D04, D05, D06, D07, D08, D09, D10, D11, D12
	U05	M05	E01, E02, E03, E04, E05, E06, E07, E08
	U06	M06	F01, F02, F03, F04, F05, F06, F07, F08, F09
	U07	M07	G01, G02, G03, G04, G05, G06, G07, G08, G09, G10, G11, G12
	U08	M08	H01, H02, H03, H04, H05, H06, H07, H08, H09, H10, H11, H12, H13
	U09	M09	I01, I02
		M12	I03, I04, I05, I06, I07
	U10	M10	J01, J02, J03, J04, J05, J06, J07, J08, J09
	U11	M11	K01, K02, K03, K04, K05, K06, K07, K08, K09
	BU1, BU2, BU3, BU4	BM1, BM2, BM3, BM4	

Module Title	Maintaining files & folders
Module Code	K72S004M01
Module Type	Core module
Duration (Hrs.)	24 Hours (T-18, P06)
Learning Outcomes	After completion of this module the trainee will be able to: • Identify components of Windows desktop environment. • Modify operating system configuration of a computer. • Handle files and folders according to the given requirement • Perform system maintenance to optimize computer performance.
Learning Content	Theory: • Setting up a computer for use • Basic components of an operating system and its user interface • How to identify configuration of an OS and modify its settings • File systems and controlling files and folders. • How to optimize a computer performance and available tools to achieve it.
	Practical: • Identify system configuration and make necessary changes in settings for a given scenario. • Create user accounts with different types of control. • Create folder structure and modify the structure and its attributes according to the given instructions. • Create different types of files and change their attributes. • Use disk defragmenter tool to improve system performance
Teaching-Learning activities	• Assignments • Presentations • Illustrated talk • Demonstrations
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Performing word processing
Module Code	K72S004M02
Module Type	Core module
Duration (Hrs.)	60 Hours (T-24, P36)
Learning Outcomes	After completion of this module the trainee will be able to: perform each of the tasks mentioned here according to the relevant standards.
Learning Content	Theory: • Introduction to Word Processing ○ What is Word Processing ○ Available word processing packages ○ Features of the word processing package ▪ Elements of the interface of the word processing package • Setting up of the page setup • Available templates, usage of them, customize them and create new ones • Spelling, Grammar Grammar and Thesaurus • Text and paragraph formatting ○ Font ○ Alignment ○ Columns ○ Tab setting ○ Bullets and Numbering • Features of • Working with drawing objects • Insert Images into the document ○ Positioning of image with text wrapping • Insert charts into the document • Styles and Table of contents (TOC) • Performing mail merge • Find & replace text • Working with headers and footers • Review documents ○ Comments and Track changes • Saving documents ○ Different file formats • Print documents / envelops / labels • Security of documents ○ Passwords ○ Auto save and restoring • Uses of online word processing tools • Usage of help function for unknown activities

	Practical: • Create a CV (should include a photograph, O/L examination results) • Create a report minimum of five pages based on given scenario • Prepare a printing label sheet using the given address list
Teaching-Learning activities	• Assignments • Presentations • Demonstrations
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Preparing spread sheets
Module Code	K72S004M03
Module Type	Core module
Duration (Hrs.)	60 Hours (T-24, P36)
Learning Outcomes	After completion of this module the trainee will be able to: perform each of the tasks mentioned here according to the relevant standards.
Learning Content	Theory: • Introduction to Spreadsheet applications ○ What is Spreadsheet application ○ Available Spreadsheet application packages ○ Features of the Spreadsheet application package ▪ Elements of the interface of the Spreadsheet application package • Format work sheet ○ Number formats and their usage • Working with open windows ○ Vertical / Horizontal/ Cascade ○ Freeze panes ○ Move, Copy spreadsheets • Fill, Sort and Filter data ○ Conditional formatting • Absolute / Relative/Mix cell reference ○ Working with multiple worksheets • Apply built – in functions and formulas ○ Basic statistical concepts Mean, Median, Count, Sum, Average, Standard Deviation, Minimum, Maximum ○ Basic Logical functions IF, Nested IF • Prepare graphs and charts ○ Components of a Chart/Graphs ○ Type of Chart/Graph • Use data tools • Define a name for a cell/ cell range • Create a pivot table • Develop and customize worksheet • Set Page properties and print a spread sheet Practical: • Create a excel database including classmates’ information and present the data to class room using charts, graphs and other analytical tools • Perform a simple statistical analysis on set of Financial Data
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama

Forms of assessment	Formative Assessment + Summative Assessment
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Module Title	Preparing presentations
Module Code	K72S004M04
Module Type	Core module
Duration (Hrs.)	30 Hours (T-12, P18)
Learning Outcomes	After completion of this module the trainee will be able to: perform each of the tasks mentioned here according to the relevant standards.
Learning Content	Theory: • Introduction to Presentation Software • Creating a presentation • Apply design templates to an existing presentation • Create themes • Insert different media • Slide transition and animation • Recording a presentation • Insert action buttons and hyperlinks • Conduct the presentation • Save presentations • Print slides • Best practices in presentations Practical: • Create an interactive presentation on either of following topics ○ History of internet ○ History of computer ○ Cyber security ○ Good practices of social media usages
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Performing basic operations of cloud computing including internet and email services
Module Code	K72S004M05
Module Type	Core module
Duration (Hrs.)	48 Hours (T-24, P24)
Learning Outcomes	After completion of this module the trainee will be able to: perform each of the tasks mentioned here according to the relevant standards.
Learning Content	Theory: • Evolution of Internet, WWW and email • Function of the ISP • Methods can use to connect to Internet • Basic Protocols use in Internet and email • URL/URI, Domain name, IP Address basics • How to use a Web browser • Customizing/fine tuning the Web Browser (setting up home page, Security & Privacy settings, clearing history, cookies and browsing data) • Bookmarks/favorites • Install/uninstall plug-in/add-on/extensions for web browsers • Search engines and how the searching takes place • AI based search engines • Searching techniques • Downloading various files from Internet (documents, audios, videos, images, software etc.) • Internet ethics and safety (Plagiarism, Security, Privacy) • Introduction to emails and its uses • How to get an email account • Sending and receiving emails (using Cc, Bcc and attachments) • Replying, forwarding email messages • Creating/managing email folders • Creating/managing address book/s • How to handle spam/junk mails • Printing web page/part of a web page and email/s • What is social media and why use it? • Safety measure should take when using social media • How to make a YouTube channel • Features of document collaboration software • Features of spreadsheet collaboration software • Features of presentation collaboration software • Creating, handling and collect data using online forms • What is video conferencing? • Basic requirements for video conferencing

	• Popular video conferencing tools available • Inviting/Handling participants for video conferencing sessions • Facilities available on video conferencing tools • Basics of Cloud Computing • Services available in Cloud Computing (IaaS, PaaS, SaaS) • Cloud deployment models (Public, Private, Community, and Hybrid) • Advantages/disadvantages of Cloud computing • Popular Cloud Service Providers (e.g., Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform, IBM Cloud Services, VMware Cloud, Oracle cloud ... etc.) • How to use a cloud for collaboration? (e.g., Google Drive) Practical: • Visit to a given URL appropriately • Extract the required details/images/video using a search engine • Download required resources such as document/image/audio/video/software • Customize the web browser's setting as required • Send an email with an attachment to few people using both Cc and Bcc options • Create a collaborative document which can be edited by few people • Upload a file to a cloud storage space
Teaching-Learning activities	• Assignments • Presentation • Illustrated talk • Demonstrations
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Perform Basic database manipulation
Module Code	K72S004M06
Module Type	Core module
Duration (Hrs.)	60 Hours (T-24, P36)
Learning Outcomes	After completion of this module the trainee will be able to: Create a database with the relevant relationships, Forms, queries and reports.
Learning Content	Theory: • Introduction to Relational Database Management Systems (RDBMS) ○ What is RDBMS ○ Available RDBMS ○ Features of the RDBMS • Structure of a tables ○ Fields/columns ○ Data types ○ Field size ○ Data field Validation rules • Different keys using RDBMS ○ Primary key ○ Foreign key • Relationships among tables • Working with forms • Working with queries ○ Use of aggregate functions • Working with reports ○ Use of aggregate functions • Import data from External data source/s Practical: • Design a database with minimum 3 tables with the relevant relationship relationships, and should include one Form, one query and one report.
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Computer system, network and peripherals
Module Code	K72S004M07
Module Type	Core module
Duration (Hrs.)	60 Hours (T-36, P24)
Learning Outcomes	After completion of this module the trainee will be able to: • Identify the component of the computer system • Install and configure different application software • Obtain backup of the hard disk • Describe networking concept and component • Identify Network tools • Describe the basic network environment and related services
Learning Content	Theory: • Identifying the components of the different types of computers and their functionalities (RAM Types, Motherboard form factor, Power Supply Form factor, Hard disk – HDD, SSHD, SSD, PCIe SSD, VGA / Adaptor Cards, Processors, etc.) • Basic hardware faults • Check hardware compatibility with operating system • Preparation of the computer for installation • Installing the operating system • Installing device drivers • Installing application software • Backups and restore points • Ensure standard performance • Disk cleanup and Defragmentation • Importance of power management system • Principals of Networking ○ Components and types of computer networks • basic networking concepts and technologies ○ Configure network connectivity between PCs. • types of topologies ○ purpose and characteristics of networking standards ○ LAN topologies • physical components of a network ○ purpose of physical components of a network • Ethernet standards ○ Wired and Wireless Standards • Networking Standards ○ OSI and TCP/IP data models • Identification of routers, switches and other network devices • Importance of installing anti- malware software

	• Basics of cyber threats, their effect and prevention method • Importance of Ethical usage and Legal implications of unauthorized software/data usage ○ Software Licensing 48 hours
	Practical: • Identify a basic hardware fault • Install an operating system and make sure standard performance of the computer (Group) • Configure computer with wireless network • Network three computers using switch • Create cross and straight two cables 24 hours
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Producing Print Page Layout and multimedia objects
Module Code	K72S004M08
Module Type	Core module
Duration (Hrs.)	78 Hours (T-30, P48)
Learning Outcomes	After completion of this module the trainee will be able to: perform each of the tasks mentioned here according to the relevant standards.
Learning Content	Theory: • Introduction to graphic design ○ History of graphic design ○ Principals of graphic design ○ Basic design requirements ○ Design process ○ Basic theory of color ○ Introduction to typography • Introduction to printing methods • Digital file formats (Vector/ Raster) • Introduction to raster image reproduction • Multimedia creations ○ Introduction of multimedia production methods ○ Introduction of multimedia format types ○ Introduction of latest technologies in multimedia production ○ Integration of components for multimedia production ○ Techniques of multimedia file creating ○ Multimedia file sharing in web base ○ Describing web base multimedia players ○ Describing multimedia hosting platforms
	Practical: • Basic operations of graphic design software ○ Preparing page layout in InDesign ○ Editing raster images with Photoshop ○ Editing vector images with Illustrator • Basic tonal corrections with photoshop • Creation and usage of layout templates • Develop basic design base on customer requirement • Develop multimedia creation by using video editing and motion graphic application (Mobile base)
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Presentations
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Systems Analysis and Designing
Module Code	K72S004M09
Module Type	Core Competency
Duration (Hrs.)	60 Hours (T36, P12)
Learning Outcomes	After completion of this module the trainee will be able to: • Identify the phases of the System Development Life Cycle • identify the data gathering techniques. • Model a business solution using data flow diagramming (DFD). • Specify processes using flowcharts/ pseudocodes. • Undertake object modeling using the Unified Modeling Language (UML) • Describe the importance of software testing and test plan. • Prepare system documentation
Learning Content	<u>Theory:</u> • Introduction to Software Development Approaches - Waterfall method - Agile methods • Types of requirements - Functional requirements - Non-functional requirements • Requirements gathering techniques. - interviews - observations - questionnaires - report review • Software Requirement Specification (SRS) - Purpose - Basic characteristics of SRS • Type of system design methodologies - Object Oriented Analysis and Design with UML • Use case diagram. • Class diagram - Structured System Analysis and Design (SSAD) with DFD • Level 0 (Context level diagram) • Level 1(Top level diagram) • Level 2 diagram as required. • Introduction to Software Testing Techniques - Unit testing - Integration testing - System testing - Acceptance testing. • Writing test cases Duration 48 hrs.

	<u>Practical</u> • Prepare an SRS for a given case study/scenario. • Prepare system design document using DFD/ UML for a given case study/scenario. Duration 12 hrs.
Teaching-Learning activities	• Demonstrations • Video • Presentation • Lecturers • Practical Works
Forms of Assessment	Formative Assessment + Summative Assessment

Module Title	Perform relational databases using standard SQL
Module Code	K72S004M10
Module Type	Core Competency
Duration (Hrs.)	60 Hours (T-24, P-36)
Learning Outcomes	After completion of this module the trainee will be able to: • Describe different database types • Develop Entity Relationships (ER) diagrams • Converting ER to the database • Use standard SQL statements to create tables and queries • Backup and restore database/data
Learning Content	<u>Theory:</u> • Introduction to commonly available database management systems ○ The evolution of database technology ○ Components of a DBMS ○ Advantages of computerized DBMS over manual approach • Introduction to different database types ○ Relational ○ Non-relational databases • Entity Relationships (ER) diagrams ○ ER concept & terminology ○ Entities, Relationship & Attributes ○ ER Diagrams ○ Mapping conceptual model in to relational schema • Data Normalization process and the normal forms ○ 1st Normal Form (1st NF) ○ 2nd Normal Form (2nd NF) ○ 3rd Normal Form (3rd NF) • Introduction to SQL ○ Data Definition Language (DDL) ▪ Create, Alter, Drop ▪ Constraints (Not null, Unique, Primary key, foreign key, Check, Default) ○ Data Manipulation Language (DML) • Insert, Update, Delete, Select • Sub queries • Create Views • Use of SQL function (Scaler functions and aggregated functions) • Backup and restore database /data <u>Practical:</u> • Draw ER diagram for given case study / scenario

	• Design a database with minimum 3 tables with the relevant relationships • Write suitable SQL statement to perform following tasks ○ Create tables with primary keys/foreign keys ○ Perform simple queries for given scenarios ○ Create sub queries • Perform the backup and restore operation as required
Teaching-Learning activities	• Illustrated talk • Demonstrations • Video • Presentation & Lecturers
Forms of Assessment	Formative Assessment + Summative Assessment

Module Title	Designing static Website
Module Code	K72S004M11
Module Type	Core module
Duration (Hrs.)	30 Hours (T-12, P-18)
Learning Outcomes	After completion of this module the trainee will be able to: Develop a static website using HTML and CSS following the best practices
Learning Content	Theory: • Introduction to Web Designing ○ What is a web page / website ○ Difference between Static and Dynamic web page ○ Web designing tools ○ Web designing best practices • Identify web site requirement • Develop web page layout ○ Site map ○ The navigation structures • HTML (Hypertext Markup Language) ○ Introduction to HTML ○ HTML elements ○ HTML forms ○ Use hyperlinks • CSS (Cascading Style Sheets) ○ Introduction to CSS ○ CSS syntax ○ CSS apply methods (inline, internal, external) • Embed different object/media to web page Practical: • Develop a web site minimum with 3 webpages using HTML and CSS in given case study / scenario
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Developing Desktop Applications
Module Code	K72S004M12
Module Type	Core Competency
Duration (Hrs.)	72 Hours (T-48, P-36)
Learning Outcomes	After completion of this module the trainee will be able to: • Draw conceptual diagrams • Select and acquire the appropriate software for the system • Develop software application • Identify and develop the Graphical User Interface • Prepare user guide • Deploy software package
Learning Content	<u>Theory</u> • Development software - Commonly available software and their suitability • Programming fundamentals - Introduction to Programming • Structured programming vs Object Oriented Programming - Data Types and Variables - Coding standards and naming conventions - Operators - Control structures - Functions (Built-in and user defined) - Procedures - Modules • Develop programs using IDE • Connecting a Database to the developed program - Write code for Insert/ Update/ Delete/ Select operations • Test the developed program using the test cases written • Deploy the developed software

	<u>Practical</u> • Develop software application with the database connection for a given case study/scenario • Test the above developed software application using written test cases
Teaching-Learning activities	• Illustrated talk • Demonstrations • Video • Presentation • Lecturers • Sample coding
Forms of Assessment	Formative Assessment + Summative Assessment

Module Title	Communication skills for workplace
Module Code	K72S004BM01
Module Type	Basic Module
Duration (Hrs.)	24 Hours
Learning Outcomes	After completion of this module the trainee will be able to: - communicate ideas and information at the workplace effectively
Learning Content	Theory: • Listening skills • Telephone etiquettes • Presentation skills • Verbal and non-verbal communication • Safety symbols & procedures • Basic English relevant to workplace and type of work • Organization's rules, regulations and procedures • General norms of polite conduct • Forms of Different manner of address appropriately to customers, superior and subordinates and persons in different positions • Customs and practices of different ethnic and religious groups • Gender issues relevant to the communications • Trends in related technology • Friendliness through a friendly tone, a personal question, or simply a smile, encourage coworkers to engage in open and honest communication • Respect – convey respect for others and their ideas Practical: • Role plays and dramas • Assignments 01 - Provide opportunities to trainees to present learned content • Assignments 02 – Prepare list of 250 – 300 technical / vocational terms in English used in relevant industry during the course period • Assignment 03 – Collect manufacturers guidelines/ operational instructions / specifications in relevant field, prepare a file and interpret the information
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Workshop calculation & science
Module Code	K72S004BM02
Module Type	Basic Module
Duration (Hrs)	30 Hours
Learning Outcomes	After completion of this module the trainee will be able to: • Perform basic arithmetic and trigonometric operations in the workplace when and where required. • Describe electrical characteristics related to hardware process
Learning Content	Theory • Basic Arithmetic. • Area, Volume • Basic trigonometry and angles. • Units and Measurements (Length, Volume, Speed, Time). • Conversion from metric to BS and vice versa. • Basic idea of Work, Energy, power & Pressure • Units & measurement • Basic electricity, Electricity principles, Ground connections, Introduction to voltage, Current, Resistance, Power, Energy. Clip on meter, Voltmeter, Conductors & insulators
	Practical • Practice arithmetic and trigonometric operations, • Measuring Practice
Teaching-Learning activities	• Illustrated talk • Demonstrations • Group and individual practices • Use video clips • Assignments
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Team work
Module Code	K72S004BM03
Module Type	Basic Module
Duration (Hrs)	6 Hours
Learning Outcomes	At the end of this module student should be able to; • Contribute positively to the work in team environment • Work effectively with others in a socially diverse environment • Respect and understand the views of others • Give, receive and act upon feedback • Identify and describe own roles and roles of others
Learning Content	Theory: • Simple concepts of team behavior • Elementary group dynamics • Basic knowledge in social and demographic structures Practical • Identify and establish team purpose, team member's roles and responsibilities. • Coach and motivate team. • Assignment 01- Form a group for workshop cleaning, machine maintenance, waste management and give the feedback on group effort • Assignment 02- Arrange sports activities / social events
Teaching-Learning activities	• Assignments • Lecturers • role plays • case studies • team building activities • coaching and mentoring
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Occupational safety & health and environmental aspects
Module Code	K72S004BM04
Module Type	Basic Module
Duration (Hrs.)	18 Hours
Learning Outcomes	After completion of this module the trainee will be able to: • Describe importance of safety • Use personal safety gears • Arrange the place to minimize health risk • Give first aid
Learning Content	Theory • Importance of safety and general precautions to be observed in the workshop. • Personal safety gears available to protect body; how and when they use • safety signs - for danger, warning, caution & personal safety message • Safe handling of, Fire extinguishers used for Different types of fire. safe handling and Periodic testing of lifting equipment, Energy Conservation • Types of waste materials • Environmental rules and regulations on disposal of waste (solid & gases) and noise into environment • Standard treatment and disposal methods of waste (solid & gases) and noise • Method of reuse of waste (solid & gases) and scraps Practical • Practical related to Safety and Health, • Importance of maintenance and cleanliness of workshop. • Demonstration on First aid and Fire safety, • Use personal safety gears • Arrange the place of work to minimize the health risks. • Arrange the place of work in order to make it possible to work in the most ergonomic way. • Arrange exhaust system • Give first aid. • Use of fire extinguishers. • Demonstration on safe handling and Periodic testing of lifting equipment • Energy saving Tips • Collect details of waste (solid & gases) • process and prepare a brief report with proposal of appropriate disposing/ reusing methods of them

Teaching-Learning activities	• Illustrated talk • Use video clips • Demonstrations • Assignments
Forms of assessment	Formative Assessment + Summative Assessment

Assessment Guide:**Forms of assessment**

Assessment shall be based on evidence collected through workplace performance or a combination of evidence collected through training and work place performance.

Assessment context

This unit may be assessed on the job, off the job or a combination of on and off the job. The unit may be assessed individually.

Assessment conditions

The candidate will have access to:

- all tools, equipment, material and documentation required.

The candidate will be permitted to refer the following documents:

- material, paint and polish safety data sheets
- health and safety regulations
- product data sheet (PDS)
- company specification sheets
- manufacturer's instructional brochures and manuals

The candidate will be required to:

- orally or by other methods of communication, answer questions asked by the assessor.
- identify superiors and clients who can be approached for the collection of competency evidence where appropriate.
- present evidence of credit for any off-job training related to this unit.

Assessors must be satisfied that the candidate can competently and consistently perform all elements of the unit as specified by the criteria and that he/she possess the required underpinning knowledge.

Special notes

During assessment, the candidate will;

- demonstrate safe work practices at all times
- communicate information about processes, events or tasks being undertaken to ensure a safe and efficient working environment.
- perform all tasks in accordance with standard operating procedures.
- perform all tasks to specifications
- take responsibility for the accuracy of his/her own work
- use accepted methods for calculation.

Tasks involved will be completed within reasonable time frames relating to typical workplace activities.

Resources required for assessment

These include material, tools and equipment listed within this unit

Tools, equipment and material required for the training programme

(Approximately 15 students)

No	Item	Quantity
01	Multimedia capable computer	15
02	Internet facility	For 15 computers
03	Standard operating system software	Install for 15 computers
05	Software applications for relevant units	Install for 15 computers
06	Printer	01
07	Printing materials	Adequate quantity
08	Multimedia projector	01
09	Scanner	03 01
10	Microphone	15
11	Webcam	15
12	Computer technician's tool kit	Set 01
13	Network tool kit	Set 01
14	Network devices	Adequate quantity
15	Cables and tools	Adequate quantity
16	Cable testers	Adequate quantity
17	Crimping tools	Adequate quantity
18	Connectors	Adequate quantity
20	Stationary	Adequate quantity

NATIONAL COMPETENCY STANDARD & COMPETENCY BASED TRAINING CURRICULUM For INFORMATION AND COMMUNICATION TECHNOLOGY TECHNICIAN NVQ Level 02, 03 and 04 (Competency Standards Code: K72S004)	
1. Endorsement Date:31.10.2023	2. Date of Review : 31.10.2026
3. Purpose of the Qualification	To certify that the holder of this qualification has acquired the competencies contained in the units.
4. Regulations for the Qualification/s	The holder should have been assessed by a licensed assessor and found competent in the units and certified by the TVEC
5. Qualification Codes & Packages	- National Certificate at NVQ Level 2 in the occupation of Data Entry Operator will be awarded to those who are competent in Unit01 + Unit02 + Unit05 + BU01 + BU02 + BU03 + BU04 – K72S004Q1L2 - National Certificate at NVQ Level 3 in the occupation of Computer Application Assistant will be awarded to those who are competent in Unit01 + Unit02 + Unit03 + Unit04 + Unit05 + Unit06 + Unit07 + BU01 + BU02 + BU03 + BU04 – K72S004Q2L3 - National Certificate at NVQ Level 4 in the occupation of information and Communication Technology Technician will be awarded to those who are competent in units Unit01 + Unit02 + Unit03 + Unit04 + Unit05 + Unit06 + Unit07 + Unit08 + Unit09 +Unit10 +Unit11 + BU01 + BU02 + BU03 + BU04 – K72S004Q3L4
6. Prerequisites	The candidate should have been found competent in Basic competencies to work to be eligible for the award of this qualification. The candidate may concurrently apply for basic competencies to work and this qualification
7. Accreditation Requirement	The qualifications shall be offered in compliance with the accreditation requirements of the TVEC as stipulated in the National Vocational Qualifications framework of Sri Lanka.
8. Certification	TVEC shall certify the qualifications in terms of the regulation.
9. Transition arrangements	The competency-based assessments shall be undertaken by the registered assessors until TVEC arranges to issue licenses to the assessors
10. Requirement of conducting Knowledge Assessment	Required
11. Contact for Comments	Chairman, National Apprentice & Industrial Training Authority, 971, Sri Jayewardenepura Mawatha, Welikada, Rajagiriya. Director General, Tertiary & Vocational Education Commission, 354/2, Nipunatha Piyasa, Elvitigala Mawatha, Colombo 05.

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